UNITED STATES DEPARTMENT OF COMMERCE, Charles Sawyer, Secretary
NATIONAL BUREAU OF STANDARDS, A. V. Astin, Director

ATOMIC ENERGY LEVELS

As Derived From the Analyses of Optical Spectra

Volume II

The Spectra of Chromium, Manganese, Iron, Cobalt, Nickel, Copper, Zinc, Gallium, Germanium, Arsenic, Selenium, Bromine, Krypton, Rubidium, Strontium, Yttrium, Zirconium, and Niobium

By CHARLOTTE E. MOORE



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RUBIDIUM

Rb I

37 electrons $Z=37$

Ground state $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 5s\ {}^2S_{0\frac{1}{2}}$

$5s\ {}^2S_{0\frac{1}{2}}$ Rb^{85} **33691.02±0.03** cm^{-1} I. P. 4.176 volts

$5s\ {}^2S_{0\frac{1}{2}}$ Rb^{87} **33691.10±0.03** cm^{-1} I. P. 4.176 volts

The well known series of Rb I reported in the classical publications of Paschen-Götze and Fowler have recently been extended. Kratz has observed in absorption the principal series ($np\ {}^2P°$) to $n=77$.

Mack has furnished in advance of publication his observations of forbidden absorption lines, all of which are combinations of the ground term, $5s\ {}^2S$, with higher series members $nd\ {}^2D$ and $ns\ {}^2S$. The series called $nf\ {}^2F°$ in the table is labeled by Mack $n\ {}^2Q$ to indicate that the terms are $nf\ {}^2F°$, $ng\ {}^2G$, $nh\ {}^2H°$, etc.

Ramb has observed with the interferometer 33 lines between 5169.651 A and 10075.711 A. The inverted terms of the Bergmann series have been discussed in detail by Meissner and Masaki. From these and other papers Mack has prepared the list of terms as far as $12s\ {}^2S$, and furnished his manuscript for inclusion here. He notes that the values of the terms $6s\ {}^2S$, $4d\ {}^2D$, 4 to $8f\ {}^2F°$, 5, $6g\ {}^2G$, and $6h\ {}^2H°$ need further adjustment from more accurate wave lengths. He suggests, also, the following improved values of intervals:

Term	Interval	Term	Interval
$5p\ {}^2P°$	237. 598	$7d\ {}^2D$	1. 507
$5d\ {}^2D$	2. 960	$8d\ {}^2D$	1. 013
$6d\ {}^2D$	2. 262	$9d\ {}^2D$	0. 697

Brackets denote that the tabular entries have been derived from series calculations.

Kratz derives the limit 33690.96 ±0.03 cm^{-1}, from the long $np\ {}^2P°$ series, and calculates the limit of the lowest hyperfine structure sublevel as 33691.02 ±0.03. Mack confirms this calculation for Rb^{85} and adds the value quoted above for Rb^{87}.

Beutler has observed in absorption 39 lines between 594.61 A and 809.72 A, and classified all but 3 as transitions from the ground term to levels above the ionization limit. His rounded-off wave numbers have been entered in the table. These levels have as limits, two higher terms ${}^3P°$ and ${}^1P°$ in Rb II. The miscellaneous levels have been assigned numbers by the writer. The double entries of J for these levels (unlike those for unresolved terms) indicate that the existing data are insufficient to distinguish which value of J is correct.

The Quadratic Zeeman Effect in the principal series of Rb I is discussed by Harting and Klinkenberg.

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Config.	Desig.	J	Level	Interval
$4p^6(^1S)5s$	$5s\ ^2S$	0½	0. 00	
$4p^6(^1S)5p$	$5p\ ^2P°$	0½	*12578. 96*	237. 60
		1½	*12816. 56*	
$4p^6(^1S)4d$	$4d\ ^2D$	2½	19355. 01	−0. 44
		1½	19355. 45	
$4p^6(^1S)6s$	$6s\ ^2S$	0½	20133. 6	
$4p^6(^1S)6p$	$6p\ ^2P°$	0½	*23715. 19*	77. 50
		1½	*23792. 69*	
$4p^6(^1S)5d$	$5d\ ^2D$	1½	25700. 56	2. 96
		2½	25703. 52	
$4p^6(^1S)7s$	$7s\ ^2S$	0½	26311. 46	
$4p^6(^1S)4f$	$4f\ ^2F°$	2½, 3½	*26791. 98*	
$4p^6(^1S)7p$	$7p\ ^2P°$	0½	*27835. 05*	35. 09
		1½	*27870. 14*	
$4p^6(^1S)6d$	$6d\ ^2D$	1½	28687. 15	2. 26
		2½	28689. 41	
$4p^6(^1S)8s$	$8s\ ^2S$	0½	29046. 84	
$4p^6(^1S)5f$	$5f\ ^2F°$	3½	*29277. 58*	−0. 01
		2½	*29277. 59*	
$4p^6(^1S)5g$	$5g\ ^2G$	3½, 4½	29297. 6	
$4p^6(^1S)8p$	$8p\ ^2P°$	0½	*29834. 96*	18. 86
		1½	*29853. 82*	
$4p^6(^1S)7d$	$7d\ ^2D$	1½	30280. 18	1. 51
		2½	30281. 69	
$4p^6(^1S)9s$	$9s\ ^2S$	0½	30499. 06	
$4p^6(^1S)6f$	$6f\ ^2F°$	3½	*30627. 77*	−0. 01
		2½	*30627. 78*	
$4p^6(^1S)6g$	$6g\ ^2G$	3½, 4½	30636. 9	
$4p^6(^1S)6h$	$6h\ ^2H°$	4½, 5½	*30643. 5*	
$4p^6(^1S)9p$	$9p\ ^2P°$	0½	*30958. 94*	11. 28
		1½	*30970. 22*	
$4p^6(^1S)8d$	$8d\ ^2D$	1½	31221. 47	1. 01
		2½	31222. 48	
$4p^6(^1S)10s$	$10s\ ^2S$	0½	31362. 36	
$4p^6(^1S)7f$	$7f\ ^2F°$	3½	*31441. 52*	−0. 01
		2½	*31441. 53*	
$4p^6(^1S)10p$	$10p\ ^2P°$	0½	*31653. 88*	7. 31
		1½	*31661. 19*	
$4p^6(^1S)9d$	$9d\ ^2D$	1½	31821. 88	0. 70
		2½	31822. 58	
$4p^6(^1S)11s$	$11s\ ^2S$	0½	31917. 25	
$4p^6(^1S)8f$	$8f\ ^2F°$	2½, 3½	*31969. 43*	
$4p^6(^1S)11p$	$11p\ ^2P°$	0½	*32113. 58*	4. 97
		1½	*32118. 55*	
$4p^6(^1S)10d$	$10d\ ^2D$	1½	[32227. 68]	[0. 48]
		2½	32228. 16	
$4p^6(^1S)12s$	$12s\ ^2S$	0½	[32295. 04]	
$4p^6(^1S)12p$	$12p\ ^2P°$	0½	*32433. 50*	3. 54
		1½	*32437. 04*	

Config.	Desig.	J	Level	Interval
$4p^6(^1S)11d$	$11d\ ^2D$	1½	32514. 63	0. 30
		2½	32514. 93	
$4p^6(^1S)13p$	$13p\ ^2P°$	0½	*32665. 03*	2. 60
		1½	*32667. 63*	
$4p^6(^1S)12d$	$12d\ ^2D$	1½	32724. 68	0. 25
		2½	32724. 93	
$4p^6(^1S)14p$	$14p\ ^2P°$	0½	*32838. 02*	2. 00
		1½	*32840. 02*	
$4p^6(^1S)13d$	$13d\ ^2D$	1½	32883. 21	0. 20
		2½	32883. 41	
$4p^6(^1S)15p$	$15p\ ^2P°$	0½	*32970. 66*	1. 53
		1½	*32972. 19*	
$4p^6(^1S)14d$	$14d\ ^2D$	1½, 2½	33005. 84	
$4p^6(^1S)16p$	$16p\ ^2P°$	0½	*33074. 59*	1. 24
		1½	*33075. 83*	
$4p^6(^1S)15d$	$15d\ ^2D$	1½, 2½	33102. 39	
$4p^6(^1S)17p$	$17p\ ^2P°$	0½	*33157. 54*	1. 00
		1½	*33158. 54*	
$4p^6(^1S)16d$	$16d\ ^2D$	1½, 2½	33180. 03	
$4p^6(^1S)18p$	$18p\ ^2P°$	0½	*33224. 83*	0. 84
		1½	*33225. 67*	
$4p^6(^1S)17d$	$17d\ ^2D$	1½, 2½	33244. 23	
$4p^6(^1S)19p$	$19p\ ^2P°$	0½	*33280. 13*	0. 68
		1½	*33280. 81*	
$4p^6(^1S)18d$	$18d\ ^2D$	1½, 2½	33295. 28	
$4p^6(^1S)20p$	$20p\ ^2P°$	0½	*33326. 13*	0. 57
		1½	*33326. 70*	
$4p^6(^1S)19d$	$19d\ ^2D$	1½, 2½	33338. 80	
$4p^6(^1S)21p$	$21p\ ^2P°$	0½	*33364. 81*	0. 48
		1½	*33365. 29*	
$4p^6(^1S)20d$	$20d\ ^2D$	1½, 2½	33375. 4	
$4p^6(^1S)22p$	$22p\ ^2P°$	0½	*33397. 66*	0. 43
		1½	*33398. 09*	
$4p^6(^1S)21d$	$21d\ ^2D$	1½, 2½	33406. 86	
$4p^6(^1S)23p$	$23p\ ^2P°$	0½	*33425. 76*	0. 39
		1½	*33426. 15*	
$4p^6(^1S)22d$	$22d\ ^2D$	1½, 2½	33433. 88	
$4p^6(^1S)24p$	$24p\ ^2P°$	0½	*33450. 03*	0. 35
		1½	*33450. 38*	
$4p^6(^1S)23d$	$23d\ ^2D$	1½, 2½	33457. 06	
$4p^6(^1S)25p$	$25p\ ^2P°$	0½	*33471. 11*	0. 32
		1½	*33471. 43*	
$4p^6(^1S)24d$	$24d\ ^2D$	1½, 2½	33477. 0	
$4p^6(^1S)26p$	$26p\ ^2P°$	0½	*33489. 53*	0. 26
		1½	*33489. 79*	
$4p^6(^1S)25d$	$25d\ ^2D$	1½, 2½	33494. 8	
$4p^6(^1S)27p$	$27p\ ^2P°$	0½, 1½	*33505. 92*	
$4p^6(^1S)26d$	$26d\ ^2D$	1½, 2½	33510. 1	
$4p^6(^1S)28p$	$28p\ ^2P°$	0½, 1½	*33520. 22*	

Rb I—Continued

Config.	Desig.	J	Level	Interval
$4p^6(^1S)27d$	$27d\ ^2D$	1½, 2½	33524. 01	
$4p^6(^1S)29p$	$29p\ ^2P°$	0½, 1½	*33532. 91*	
$4p^6(^1S)28d$	$28d\ ^2D$	1½, 2½	33535. 9	
$4p^6(^1S)30p$	$30p\ ^2P°$	0½, 1½	*33544. 26*	
$4p^6(^1S)29d$	$29d\ ^2D$	1½, 2½	33547. 3	
$4p^6(^1S)31p$	$31p\ ^2P°$	0½, 1½	*33554. 47*	
$4p^6(^1S)30d$	$30d\ ^2D$	1½, 2½	33557. 02	
$4p^6(^1S)32s$	$32s\ ^2S$	0½	33559. 2	
$4p^6(^1S)29f$	$29f\ ^2F°$	2½, etc.	*33559. 9*	
$4p^6(^1S)32p$	$32p\ ^2P°$	0½, 1½	*33563. 57*	
$4p^6(^1S)31d$	$31d\ ^2D$	1½, 2½	33565. 95	
$4p^6(^1S)33s$	$33s\ ^2S$	0½	33567. 7	
$4p^6(^1S)30f$	$30f\ ^2F°$	2½, etc.	*33568. 5*	
$4p^6(^1S)33p$	$33p\ ^2P°$	0½, 1½	*33571. 85*	
$4p^6(^1S)32d$	$32d\ ^2D$	1½, 2½	33574. 06	
$4p^6(^1S)34s$	$34s\ ^2S$	0½	[33575. 6]	
$4p^6(^1S)31f$	$31f\ ^2F°$	2½, etc.	[*33576. 7*]	
$4p^6(^1S)34p$	$34p\ ^2P°$	0½, 1½	*33579. 34*	
$4p^6(^1S)33d$	$33d\ ^2D$	1½, 2½	33581. 21	
$4p^6(^1S)35s$	$35s\ ^2S$	0½	33582. 7	
$4p^6(^1S)32f$	$32f\ ^2F°$	2½, etc.	*33583. 8*	
$4p^6(^1S)35p$	$35p\ ^2P°$	0½, 1½	*33586. 11*	
$4p^6(^1S)34d$	$34d\ ^2D$	1½, 2½	33587. 87	
$4p^6(^1S)36s$	$36s\ ^2S$	0½	33589. 09	
$4p^6(^1S)33f$	$33f\ ^2F°$	2½, etc.	*33589. 93*	
$4p^6(^1S)36p$	$36p\ ^2P°$	0½, 1½	*33592. 31*	
$4p^6(^1S)35d$	$35d\ ^2D$	1½, 2½	33593. 83	
$4p^6(^1S)37s$	$37s\ ^2S$	0½	33595. 02	
$4p^6(^1S)34f$	$34f\ ^2F°$	2½, etc.	*33595. 9*	
$4p^6(^1S)37p$	$37p\ ^2P°$	0½, 1½	*33597. 96*	
$4p^6(^1S)36d$	$36d\ ^2D$	1½, 2½	33599. 42	
$4p^6(^1S)38s$	$38s\ ^2S$	0½	33600. 57	
$4p^6(^1S)35f$	$35f\ ^2F°$	2½, etc.	*33601. 44*	
$4p^6(^1S)38p$	$38p\ ^2P°$	0½, 1½	*33603. 14*	
$4p^6(^1S)37d$	$37d\ ^2D$	1½, 2½	33604. 43	
$4p^6(^1S)39s$	$39s\ ^2S$	0½	33605. 26	
$4p^6(^1S)36f$	$36f\ ^2F°$	2½, etc.	*33605. 97*	
$4p^6(^1S)39p$	$39p\ ^2P°$	0½, 1½	*33607. 96*	
$4p^6(^1S)38d$	$38d\ ^2D$	1½, 2½	33609. 08	
$4p^6(^1S)40s$	$40s\ ^2S$	0½	33610. 09	
$4p^6(^1S)37f$	$37f\ ^2F°$	2½, etc.	*33610. 41*	

Rb I—Continued

Config.	Desig.	J	Level	Interval
$4p^6(^1S)40p$	$40p\ ^2P°$	0½, 1½	*33612. 34*	
$4p^6(^1S)39d$	$39d\ ^2D$	1½, 2½	33613. 38	
$4p^6(^1S)41s$	$41s\ ^2S$	0½	33614. 23	
$4p^6(^1S)38f$	$38f\ ^2F°$	2½, etc.	*33614. 55*	
$4p^6(^1S)41p$	$41p\ ^2P°$	0½, 1½	*33616. 38*	
$4p^6(^1S)40d$	$40d\ ^2D$	1½, 2½	33617. 34	
$4p^6(^1S)42s$	$42s\ ^2S$	0½	33618. 26	
$4p^6(^1S)39f$	$39f\ ^2F°$	2½, etc.	*33619. 5*	
$4p^6(^1S)42p$	$42p\ ^2P°$	0½, 1½	*33620. 15*	
$4p^6(^1S)41d$	$41d\ ^2D$	1½, 2½	33621. 00	
$4p^6(^1S)43s$	$43s\ ^2S$	0½	33621. 62	
$4p^6(^1S)40f$	$40f\ ^2F°$	2½, etc.	*33622. 05*	
$4p^6(^1S)43p$	$43p\ ^2P°$	0½, 1½	*33623. 61*	
$4p^6(^1S)42d$	$42d\ ^2D$	1½, 2½	33624. 41	
$4p^6(^1S)44s$	$44s\ ^2S$	0½	33625. 08	
$4p^6(^1S)41f$	$41f\ ^2F°$	2½, etc.	*33625. 56*	
$4p^6(^1S)44p$	$44p\ ^2P°$	0½, 1½	*33626. 83*	
$4p^6(^1S)43d$	$43d\ ^2D$	1½, 2½	33627. 55	
$4p^6(^1S)45s$	$45s\ ^2S$	0½	33628. 23	
$4p^6(^1S)42f$	$42f\ ^2F°$	2½, etc.	*33628. 76*	
$4p^6(^1S)45p$	$45p\ ^2P°$	0½, 1½	*33629. 80*	
$4p^6(^1S)44d$	$44d\ ^2D$	1½, 2½	33630. 51	
$4p^6(^1S)46s$	$46s\ ^2S$	0½	33631. 13	
$4p^6(^1S)43f$	$43f\ ^2F°$	2½, etc.	[*33631. 6*]	
$4p^6(^1S)46p$	$46p\ ^2P°$	0½, 1½	*33632. 60*	
$4p^6(^1S)45d$	$45d\ ^2D$	1½, 2½	33633. 23	
$4p^6(^1S)47s$	$47s\ ^2S$	0½	33633. 93	
$4p^6(^1S)44f$	$44f\ ^2F°$	2½, etc.	[*33634. 28*]	
$4p^6(^1S)47p$	$47p\ ^2P°$	0½, 1½	*33635. 21*	
$4p^6(^1S)46d$	$46d\ ^2D$	1½, 2½	33635. 83	
$4p^6(^1S)48s$	$48s\ ^2S$	0½	33636. 33	
$4p^6(^1S)45f$	$45f\ ^2F°$	2½, etc.	*33636. 86*	
$4p^6(^1S)48p$	$48p\ ^2P°$	0½, 1½	*33637. 63*	
$4p^6(^1S)47d$	$47d\ ^2D$	1½, 2½	33638. 17	
$4p^6(^1S)49s$	$49s\ ^2S$	0½	33638. 80	
$4p^6(^1S)46f$	$46f\ ^2F°$	2½, etc.	[*33639. 10*]	
$4p^6(^1S)49p$	$49p\ ^2P°$	0½, 1½	*33639. 89*	
$4p^6(^1S)48d$	$48d\ ^2D$	1½, 2½	33640. 46	
$4p^6(^1S)50s$	$50s\ ^2S$	0½	33640. 95	
$4p^6(^1S)47f$	$47f\ ^2F°$	2½, etc.	*33641. 32*	
$4p^6(^1S)50p$	$50p\ ^2P°$	0½, 1½	*33642. 01*	

Rb I—Continued

Config.	Desig.	J	Level	Interval
$4p^6(^1S)49d$	$49d\ ^2D$	1½, 2½	33642. 7	
$4p^6(^1S)51s$	$51s\ ^2S$	0½	33643. 98	
$4p^6(^1S)51p$	$51p\ ^2P°$	0½, 1½	*33644. 02*	
$4p^6(^1S)50d$	$50d\ ^2D$	1½, 2½	33644. 5	
$4p^6(^1S)52s$	$52s\ ^2S$	0½	33644. 95	
$4p^6(^1S)52p$	$52p\ ^2P°$	0½, 1½	*33645. 96*	
$4p^6(^1S)51d$	$51d\ ^2D$	1½, 2½	33646. 54	
$4p^6(^1S)53s$	$53s\ ^2S$	0½	33646. 84?	
$4p^6(^1S)53p$	$53p\ ^2P°$	0½, 1½	*33647. 72*	
$4p^6(^1S)52d$	$52d\ ^2D$	1½, 2½	33648. 29	
$4p^6(^1S)54p$	$54p\ ^2P°$	0½, 1½	*33649. 33*	
$4p^6(^1S)53d$	$53d\ ^2D$	1½, 2½	33650. 03	
$4p^6(^1S)55p$	$55p\ ^2P°$	0½, 1½	*33650. 95*	
$4p^6(^1S)54d$	$54d\ ^2D$	1½, 2½	33651. 36	
$4p^6(^1S)56p$	$56p\ ^2P°$	0½, 1½	*33652. 41*	
$4p^6(^1S)57p$	$57p\ ^2P°$	0½, 1½	*33653. 80*	
$4p^6(^1S)58p$	$58p\ ^2P°$	0½, 1½	*33655. 11*	
$4p^6(^1S)59p$	$59p\ ^2P°$	0½, 1½	*33656. 39*	
$4p^6(^1S)60p$	$60p\ ^2P°$	0½, 1½	*33657. 55*	
$4p^6(^1S)61p$	$61p\ ^2P°$	0½, 1½	*33658. 69*	
$4p^6(^1S)62p$	$62p\ ^2P°$	0½, 1½	*33659. 80*	
$4p^6(^1S)63p$	$63p\ ^2P°$	0½, 1½	*33660. 82*	
$4p^6(^1S)64p$	$64p\ ^2P°$	0½, 1½	*33661. 81*	
$4p^6(^1S)65p$	$65p\ ^2P°$	0½, 1½	*33662. 73*	
$4p^6(^1S)66p$	$66p\ ^2P°$	0½, 1½	*33663. 63*	
$4p^6(^1S)67p$	$67p\ ^2P°$	0½, 1½	*33664. 48*	
$4p^6(^1S)68p$	$68p\ ^2P°$	0½, 1½	*33665. 28*	
$4p^6(^1S)69p$	$69p\ ^2P°$	0½, 1½	*33666. 04*	
$4p^6(^1S)70p$	$70p\ ^2P°$	0½, 1½	*33666. 78*	
$4p^6(^1S)71p$	$71p\ ^2P°$	0½, 1½	*33667. 49*	
$4p^6(^1S)72p$	$72p\ ^2P°$	0½, 1½	*33668. 11*	
$4p^6(^1S)73p$	$73p\ ^2P°$	0½, 1½	*33668. 79*	
$4p^6(^1S)74p$	$74p\ ^2P°$	0½, 1½	*33669. 36*	
$4p^6(^1S)75p$	$75p\ ^2P°$	0½, 1½	*33670. 01*	
$4p^6(^1S)76p$	$76p\ ^2P°$	0½, 1½	*33670. 61*	
$4p^6(^1S)77p$	$77p\ ^2P°$	0½, 1½	*33671. 07*	
Rb II ($^1S_0$)	***Limit***	-------	**33691. 02**	
$4p^5(^3P^\circ_2)5s^2$	$5s^2\ ^2P°$	1½	*123500*	
$4p^5(^1P^\circ_1)5s^2$		0½	*130320*	−6820
$4p^5\ 5s(^3P^\circ_2)4d$	$4d\ \ 1°$	0½, 1½	*138780*	

Rb I—Continued

Config.	Desig.	J	Level	Interval
$4p^5\ 5s(^3P^\circ_1)4d$	$4d\ \ 2°$	0½, 1½	*139910*	
$4p^5\ 5s(^3P^\circ_2)4d$	$4d\ \ 3°$	0½, 1½	*145540*	
$4p^5\ 5s(^3P^\circ_2)4d$	$4d\ \ 4°$	0½, 1½	*146050*	
$4p^5\ 5s(^3P^\circ_1)4d$	$4d\ \ 5°$	0½, 1½	*146780*	
$4p^5\ 5s(^3P^\circ_2)4d$	$4d\ \ 6°$	0½, 1½	*147330*	
$4p^5\ 5s(^3P^\circ_0)4d$	$4d\ \ 7°$	0½, 1½	*147580*	
$4p^5\ 5s(^3P^\circ_1)4d$	$4d\ \ 8°$	0½, 1½	*148910*	
$4p^5\ 5s(^3P^\circ_2)6s$	$6s\ ^2P°$	1½	*151760*	
$4p^5\ 5s(^3P^\circ_1)6s$		0½?	*152980*	−1220
$4p^5\ 5s(^3P°)6s$	$6s\ ^4P°$	2½		
$4p^5\ 5s(^3P^\circ_1)6s$		1½?	*152380*	
$4p^5\ 5s(^3P^\circ_0)6s$		0½?	*158520*	−6140
$4p^5\ 5s(^1P^\circ_1)4d$	$4d'\ \ 9°$	0½, 1½	*152690*	
$4p^5\ 5s(^1P^\circ_1)4d$	$4d'\ 10°$	0½, 1½	*154040*	
$4p^5\ 5s(^3P^\circ_2)5d$	$5d\ 11°$	0½, 1½	*154260*	
$4p^5\ 5s(^1P^\circ_1)4d?$	$4d'\ 12°$	0½, 1½	*154950*	
$4p^5\ 5s(^3P^\circ_1)5d$	$5d\ 13°$	0½, 1½	*156310*	
$4p^5\ 5s(^3P^\circ_2)5d$	$5d\ 14°$	0½, 1½	*156980*	
$4p^5\ 5s(^3P^\circ_2)5d$	$5d\ 15°$	0½, 1½	*157380*	
$4p^5\ 5s(^3P^\circ_1)5d$	$5d\ 16°$	0½, 1½	*157750*	
$4p^5\ 5s(^3P^\circ_1)5d$	$5d\ 17°$	0½, 1½	*159070*	
$4p^5\ 5s(^1P^\circ_1)6s$	$6s'\ ^2P°$	0½	*159210*	
		1½	*160990*	1780
$4p^5\ 5s(^3P^\circ_2)7s$	$7s\ ^2P°$	1½	*159210*	
		0½		
$4p^5\ 5s(^3P°)7s$	$7s\ ^4P°$	2½		
$4p^5\ 5s(^3P^\circ_1)7s?$		1½?	*160060*	
$4p^5\ 5s(^3P^\circ_0)7s?$		0½?	*166830*	−6770
$4p^5\ 5s(^3P^\circ_2)6d$	$6d\ 18°$	0½, 1½	*161210*	
$4p^5\ 5s(^3P^\circ_2)6d$	$6d\ 19°$	0½, 1½	*161450*	
$4p^5\ 5s(^3P^\circ_1)6d$	$6d\ 20°$	0½, 1½	*162820*	
$4p^5\ 5s(^3P^\circ_2)7d$	$7d\ 21°$	0½, 1½	*162820*	
$4p^5\ 5s(^3P^\circ_0)5d$	$5d\ 22°$	0½, 1½	*163380*	
$4p^5\ 5s(^3P°)8s$	$8s\ ^4P°$	2½		
$4p^5\ 5s(^3P^\circ_1)8s$		1½?	*163500*	
		0½		
$4p^5\ 5s(^1P^\circ_1)5d$	$5d'\ 23°$	0½, 1½	*164700*	
$4p^5\ 5s(^1P^\circ_1)5d$	$5d'\ 24°$	0½, 1½	*165080*	
$4p^5\ 5s(^1P^\circ_1)5d$	$5d'\ 25°$	0½, 1½	*166060*	
$4p^5\ 5s(^1P^\circ_1)7s$	$7s'\ ^2P°$	0½?	*168180*	
		1½		
Rb II ($5s\ ^3P^\circ_2$)	*Limit*	----------	*167040*	
Rb II ($5s\ ^3P^\circ_1$)	*Limit*	----------	*168570*	
Rb II ($5s\ ^3P^\circ_0$)	*Limit*	----------	*175570*	
Rb II ($5s\ ^1P^\circ_1$)	*Limit*	----------	*177160*	

Rb II

(Kr I sequence; 36 electrons) $Z=37$

Ground state $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ {}^1S_0$

$4p^6\ {}^1S_0$ **221852** cm^{-1} I. P. 27.5 volts

Laporte, Miller, and Sawyer have published a list of 102 classified lines extending from 697.04 A to 7698.57 A. They have utilized earlier data by Reinheimmer and Otsuka together with their own additional observations. Their limit is from the $ns\ {}^3P^\circ_{2,1}$ series ($n=5,6$) fitted to a Rydberg formula. The higher limit given in the table has been obtained by adding to their ionization limit the interval of the ground term of Rb III.

The author's notation is entered in column one of the table. The writer has provisionally assigned to the miscellaneous levels the pair-coupling notation, used in these volumes for spectra of the inert gases. These assignments require further confirmation.

REFERENCES

O. Laporte, G. R. Miller, and R. A. Sawyer, Phys. Rev. **38,** 843 (1931). (I P) (T) (C L)
J. E. Mack, Rev. Mod. Phys. **22,** No. 1, 64 (1950). (Summary hfs)

Rb II **Rb II**

Authors	Config.	Desig.	J	Level	Authors	Config.	Desig.	J	Level
$4p^6\ {}^1S_0$	$4p^6$	$4p^6\ {}^1S$	0	0. 00	$5p\ 7_1$	$4p^5(^2P^\circ_{0\frac{1}{2}})5p$	$5p'\ [1\frac{1}{2}]$	1	163929. 47
					$5p\ 8_2$			2	164972. 81
$4d\ 1^\circ_1$	$4p^5(^2P^\circ_{1\frac{1}{2}})4d$	$4d\ [0\frac{1}{2}]^\circ$	1	*126453. 53*	$5p\ 9_1$	"	$5p'\ [0\frac{1}{2}]$	1	165094. 55
$4d\ 4^\circ_0$			0	*138799. 56*	$5p\ 10_0$			0	167637. 30
$4d\ 2^\circ_2$	"	$4d\ [1\frac{1}{2}]^\circ$	2	*128693 70*					
$4d\ 5^\circ_1$			1	*140615. 18*	$6s\ {}^3P^\circ_2$	$4p^5(^2P^\circ_{1\frac{1}{2}})6s$	$6s\ [1\frac{1}{2}]^\circ$	2	*179740. 11*
					$6s\ {}^3P^\circ_1$			1	*180173. 33*
$4d\ 3^\circ_3$	"	$4d\ [2\frac{1}{2}]^\circ$	2	*136665. 21*					
			3			$4p^5(^2P^\circ_{0\frac{1}{2}})6s$	$6s'\ [0\frac{1}{2}]^\circ$	0	
					$6s\ {}^1P^\circ_1$			1	*188622. 28*
$4d\ 6^\circ_{1,2}$	$4p^5(^2P^\circ_{0\frac{1}{2}})4d$	$4d'\ [2\frac{1}{2}]^\circ$	2	*143027. 39*					
			3						
					$5d\ 1^\circ_1$	$4p^5(^2P^\circ_{1\frac{1}{2}})5d$	$5d\ [0\frac{1}{2}]^\circ$	1	*184205. 27*
$4d\ 7^\circ_2$	"	$4d'\ [1\frac{1}{2}]^\circ$	2	*143960. 94*				0	
$4d\ 8^\circ_{1,2}$			1	*145630. 06*					
					$5d\ 3^\circ_3$	"	$5d\ [3\frac{1}{2}]^\circ$	3	*185131. 62*
								4	
$5s\ {}^3P^\circ_2$	$4p^5(^2P^\circ_{1\frac{1}{2}})5s$	$5s\ [1\frac{1}{2}]^\circ$	2	*133347. 29*					
$5s\ {}^3P^\circ_1$			1	*134875. 14*	$5d\ 2^\circ_2$	"	$5d\ [1\frac{1}{2}]^\circ$	2	*184841. 65*
					$5d\ 6^\circ_1$			1	*187340. 37*
$5s\ {}^3P^\circ_0$	$4p^5(^2P^\circ_{0\frac{1}{2}})5s$	$5s'\ [0\frac{1}{2}]^\circ$	0	*141879. 24*					
$5s\ {}^1P^\circ_1$			1	*143467. 00*	$5d\ 4^\circ_2$	"	$5d\ [2\frac{1}{2}]^\circ$	2	*185622. 53*
					$5d\ 5^\circ_3$			3	*186010. 87*
$5p\ 1_1$	$4p^5(^2P^\circ_{1\frac{1}{2}})5p$	$5p\ [0\frac{1}{2}]$	1	154279. 25	$5d\ 7^\circ_1$	$4p^5(^2P^\circ_{0\frac{1}{2}})5d$	$5d'\ [1\frac{1}{2}]^\circ$	1	*189006. 27*
					$5d\ 8^\circ_{1,2}$			2	*192380. 15*
$5p\ 2_2$	"	$5p\ [2\frac{1}{2}]$	2	156742. 20					
$5p\ 3_3$			3	156900. 72					
						Rb III $(^2P^\circ_{1\frac{1}{2}})$	*Limit*	-------	**221852**
$5p\ 4_1$	"	$5p\ [1\frac{1}{2}]$	1	158156. 66					
$5p\ 5_2$			2	158717. 04		Rb III $(^2P^\circ_{0\frac{1}{2}})$	*Limit*	-------	229232
$5p\ 6_0$	"	$5p\ [0\frac{1}{2}]$	0	161205. 13					

May 1951.

Rb II OBSERVED LEVELS*

Config. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2+$	Observed Terms	
$4p^6$	$4p^6\ {}^1S$	
	ns $(n\geq 5)$	
$4p^5(^2P^\circ)nx$	$\left\{\begin{array}{l} 5,\ 6s\ {}^3P^\circ \\ 5,\ 6s\ {}^1P^\circ \end{array}\right.$	

jl-Coupling Notation

	Observed Pairs		
	ns $(n\geq 5)$	np $(n\geq 5)$	nd $(n\geq 4)$
$4p^5(^2P^\circ_{1\frac{1}{2}})nx$	5, $6s$ $[1\frac{1}{2}]^\circ$	$5p$ $[0\frac{1}{2}]$ $5p$ $[2\frac{1}{2}]$ $5p$ $[1\frac{1}{2}]$	$4,5d$ $[0\frac{1}{2}]^\circ$ $5d$ $[3\frac{1}{2}]^\circ$ $4,5d$ $[1\frac{1}{2}]^\circ$ $4,5d$ $[2\frac{1}{2}]^\circ$
$4p^5(^2P^\circ_{0\frac{1}{2}})nx'$	5, $6s'$ $[0\frac{1}{2}]^\circ$	$5p'$ $[1\frac{1}{2}]$ $5p'$ $[0\frac{1}{2}]$	$4d'$ $[2\frac{1}{2}]^\circ$ $4,5d'$ $[1\frac{1}{2}]^\circ$

*For predicted levels in the spectra of the Kr I isoelectronic sequence, see Introduction.

Rb III

(Br I sequence; 35 electrons) $Z=37$

Ground state $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^5\ {}^2P^\circ_{1\frac{1}{2}}$

$4p^5\ {}^2P^\circ_{1\frac{1}{2}}$ ***320000*** cm^{-1} I. P. 40 volts

The analysis is incomplete. Tomboulian has classified 30 lines of Rb III in the region between 482 A and 815 A as due to transitions from the ground term to 11 higher terms.

He has estimated the value of the limit by extrapolation of isoelectronic sequence data.

REFERENCE

D. H. Tomboulian, Phys. Rev. **54,** 350 (1938). (I P) (T) (C L)

Rb III **Rb III**

Config.	Desig.	J	Level	Interval	Config.	Desig.	J	Level	Interval
$4s^2\ 4p^5$	$4p^5\ {}^2P^\circ$	$1\frac{1}{2}$ $0\frac{1}{2}$	*0* *7380*	−7380	$4s^2\ 4p^4(^3P)4d$	$4d\ {}^2D$	$1\frac{1}{2}$ $2\frac{1}{2}$	173475	
$4s\ 4p^6$	$4p^6\ {}^2S$	$0\frac{1}{2}$	130036		$4s^2\ 4p^4(^3P)4d$	$4d\ {}^2P$	$0\frac{1}{2}$ $1\frac{1}{2}$	177044	
$4s^2\ 4p^4(^3P)4d$	$4d\ {}^4D$	$3\frac{1}{2}$ $2\frac{1}{2}$ $1\frac{1}{2}$ $0\frac{1}{2}$	 152353 154403	−2050	$4s^2\ 4p^4(^1D)5s$	$5s'\ {}^2D$	$1\frac{1}{2}$ $2\frac{1}{2}$	182963	
$4s^2\ 4p^4(^3P)5s$	$5s\ {}^4P$	$2\frac{1}{2}$ $1\frac{1}{2}$ $0\frac{1}{2}$	 164507 168085	−3578	$4s^2 4p^4(^1D)4d$	$4d'\ {}^2D$	$2\frac{1}{2}$ $1\frac{1}{2}$	200883	
$4s^2\ 4p^4(^3P)4d$	$4d\ {}^4P$	$0\frac{1}{2}$ $1\frac{1}{2}$ $2\frac{1}{2}$	170423 170847	424	$4s^2\ 4p^4(^1D)4d$	$4d'\ {}^2P$	$1\frac{1}{2}$ $0\frac{1}{2}$	204104	
$4s^2\ 4p^4(^3P)5s$	$5s\ {}^2P$	$1\frac{1}{2}$ $0\frac{1}{2}$	172045 175396	−3351	$4s^2\ 4p^4(^1S)5s$	$5s''\ {}^2S$	$0\frac{1}{2}$	207277	
					Rb IV $(^3P_2)$	***Limit***	------	**[*320000*]**	

January 1951.

Rb III OBSERVED TERMS*

Config. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}+$	Observed Terms	
$4s^2\ 4p^5$	$4p^5\ ^2P^\circ$	
$4s\ 4p^6$	$4p^6\ ^2S$	
	$ns\ (n\geq 5)$	$nd\ (n\geq 4)$
$4s^2\ 4p^4(^3P)nx$	$5s\ ^4P$ $5s\ ^2P$	$4d\ ^4P$ $4d\ ^4D$ $4d\ ^2P$ $4d\ ^2D$
$4s^2\ 4p^4(^1D)nx'$	$5s'\ ^2D$	$4d'\ ^2P$ $4d'\ ^2D$
$4s^2\ 4p^4(^1S)nx''$	$5s''\ ^2S$	

*For predicted terms in the spectra of the Br I isoelectronic sequence, see Vol. II, Introduction.

Rb IX

(Cu I sequence; 29 electrons) $Z=37$

Ground state $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s\ ^2S_{0\frac{1}{2}}$

$4s\ ^2S_{0\frac{1}{2}}$ cm^{-1} I. P. volts

This spectrum has not been analyzed, but Mack has observed the principal doublet, as follows:

I A	Wave No.	Desig.
583. 37	171417	$4s\ ^2S_{0\frac{1}{2}}-4p\ ^2P^i_{1\frac{1}{2}}$
628. 62	159078	$4s\ ^2S_{0\frac{1}{2}}-4p\ ^2P^i_{0\frac{1}{2}}$

REFERENCE

J. E. Mack, Phys. Rev. **38,** 193 (L) (1931). (I P) (T) (C L)

March 1950.

Rb X

(Ni I sequence; 28 electrons) Z=37

Ground state $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ ^1S_0$

$3d^{10}\ ^1S_0$ **2235100** cm^{-1} I. P. 277 volts

Edlén has observed lines from the $3d^{10}-3d^9np(n=4,5)$ and $3d^{10}-3d^9nf$ $(n=4-6)$ configurations. In figure 2 of his paper on the spectra of highly ionized atoms, some lines from each group are indicated on the photograph of vacuum spark spectra of Rb observed between 50 A and 100 A.

From preliminary unpublished measurements he has derived provisional absolute term values from the Rydberg *nf*-series. The terms in the table are from his manuscript. His unit, 10^3 cm^{-1}, has here been changed to cm^{-1}.

REFERENCES

B. Edlén, Physica **13,** No. 9, 547 (1947).
B. Edlén, letter (July 1950). (I P) (T)

Rb X **Rb X**

Config.	Desig.	J	Level	Config.	Desig.	J	Level
$3d^{10}$	$3d^{10}\ ^1S$	0	0	$3d^9(^2D)5p$	$5p\ ^3D°$	3	
						2	
$3d^9(^2D)4p$	$4p\ ^1P°$	1	*1023400*			1	*1568400*
$3d^9(^2D)4p$	$4p\ ^3D°$	3		$3d^9(^2D)5f$	$5f\ ^1P°$	1	*1783800*
		2					
		1	*1031100*	$3d^9(^2D)5f$	$5f\ ^3D°$	3	
						2	
$3d^9(^2D)4f$	$4f\ ^3P°$	2				1	*1799200*
		1	*1517200*				
		0		$3d^9(^2D)6f$	$6f\ ^1P°$	1	*1922300*
$3d^9(^2D)4f$	$4f\ ^1P°$	1	*1527700*	$3d^9(^2D)6f$	$4f\ ^3D°$	3	
						2	
$3d^9(^2D)4f$	$4f\ ^3D°$	3				1	*1935400*
		2					
		1	*1548000*				
$3d^9(^2D)5p$	$5p\ ^1P°$	1	*1559600*	Rb XI ($^2D_{2½}$)	***Limit***		**2235100**

August 1950.

Rb XI

(Co I sequence; 27 electrons) $Z=37$

Ground state $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9\ ^2D_{2½}$

$3d^9\ ^2D_{2½}$ cm^{-1} I. P. volts

Edlén has observed three lines due to the transition $3p^6\ 3d^9\ ^2D-3p^5\ 3d^{10}\ ^2P°$. From preliminary unpublished measurements he has furnished the provisional term values quoted in the table. In figure 2 of his paper on the spectra of highly ionized atoms, the lines are indicated on the photograph of the vacuum spark spectra of Rb observed between 50 A and 100 A. Another group of lines from $3d^9-3d^8\ 4p$ is also indicated in the figure.

Edlén's unit, $10^3\ cm^{-1}$, has here been changed to cm^{-1}.

REFERENCES

B. Edlén, Physica, **13,** No. 9, 547 (1947).
B. Edlén, letter (July 1950). (T)

Rb XI

Config.	Desig.	J	Level	Interval
$3p^6\ 3d^9$	$3d^9\ ^2D$	2½	0	−12300
		1½	12300	
$3p^5\ 3d^{10}$	$3d^{10}\ ^2P°$	1½	*1028700*	−72700
		0½	*1101400*	

July 1950.

Rb XII

(Fe I sequence; 26 electrons) $Z=37$

Ground state $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^8\ {}^3F_4$

$3d^8\ {}^3F_4$	cm^{-1}	I. P.	volts

This spectrum has not been analyzed, but Edlén has observed a group of lines due to terms from the $3d^8-3d^7\ 4p$ configurations. In figure 2 of his paper on the spectra of highly ionized atoms, the lines are indicated on the photograph of vacuum spark spectra of Rb observed between 50 A and 100 A.

The writer has assumed the ground state indicated above, by analogy with Ni III.

REFERENCE

B. Edlén, Physica **13,** No. 9, 547 (1947).

February 1950.

Rb XIII

(Mn I sequence; 25 electrons) $Z=37$

Ground state $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^7\ {}^4F_{4½}$

$3d^7\ {}^4F_{4½}$	cm^{-1}	I. P.	volts

This spectrum has not been analyzed, but Edlén has observed a group of lines due to terms from the $3d^7-3d^6\ 4p$ configurations. In figure 2 of his paper on the spectra of highly ionized atoms, the lines are indicated on the photograph of vacuum spark spectra of Rb observed between 50 A and 100 A.

The writer has assumed the ground state indicated above, by analogy with Co III.

REFERENCE

B. Edlén, Physica **13,** No. 9, 547 (1947).

February 1950.

Rb XIV

(Cr I sequence; 24 electrons) $Z=37$

Ground state $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ {}^5D_4$

$3d^6\ {}^5D_4$	cm^{-1}	I. P.	volts

This spectrum has not been analyzed, but Edlén has observed a group of lines due to terms from the $3d^6-3d^5\ 4p$ configurations. In figure 2 of his paper on the spectra of highly ionized atoms the lines are indicated on the photograph of vacuum spark spectra of Rb observed between 50 A and 100 A.

The writer has assumed the ground state indicated above, by analogy with Fe III.

REFERENCE

B. Edlén, Physica **13,** No. 9, 547 (1947).

February 1950.